

Platform White Paper

What GSOC Monitor is, what it can do, where its data comes from, and what its minimum viable product should be — documented from the code.

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alecbibat/gsoc-monitor @ main (f008edf)

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Executive Summary

GSOC Monitor is a working, deployed situational-awareness and incident-command platform. This paper documents it from the code: what it does, how it is built, where its data comes from, how mature each part is, and what should constitute its minimum viable product.

What it is

A browser-based 3D globe for a Global Security Operations Center that fuses live hazard feeds — weather, wildfire, smoke and air quality, earthquakes, tropical cyclones, lightning, river flooding, power outages, aircraft, ships, satellites and open-source news — into one operating picture; scans a portfolio of 31 properties in 13 groups and a seven-ship fleet against those feeds with an explainable 0–100 threat score and an optional AI-written duty-officer briefing; and, when an incident occurs, becomes a full incident-command workspace with a synchronised multi-operator record, ICS organisation chart, action log, checklists, drawn map layers, password-protected share links for stakeholders, an Incident Action Plan library, a stand-down sequence and an after-action report.

What the review found

303

CAPABILITIES
CATALOGUED

84%

PRODUCTION-READY

~90

UPSTREAM DATA FEEDS

56

VERIFIED DEFECTS

- **Breadth.** Twelve subsystems, 29 server routes, 24 map layers, seven background collectors and about ninety upstream feeds from roughly forty providers. Nearly every feed is free; the only unavoidable spend is the single Heroku dyno and its Postgres database, with optional paid tiers for dependable vessel positions, the AI briefing and photorealistic 3D imagery.
- **Depth where it matters.** The crisis workspace is engineered against lost updates on the paths that carry the record of a response — the action log and checklists are row-locked and per-entry, twelve kinds of audit events are written automatically, and share links have passwords, a 72-hour expiry, revocation and an access log. The proxy tier caches with stale-on-error and request coalescing; the globe recovers from a lost GPU context on its own.
- **Honest limits.** Everything tenant-specific — properties, fleet, aircraft, templates, briefing context — is compiled into the code. The system is single-process by design. Monitoring is pull-only with no notifications or history. Coverage is US-centric while the fleet sails internationally. Every report is produced through the browser's print dialog. One checklist template serves all 26 incident types.
- **Security posture.** A parallel adversarially-verified review found 56 defects: 9 high, 22 medium and 25 low, and no critical. The high-severity items cluster in session revocation (a deleted user's 30-day session keeps working), abuse of unauthenticated routes (arbitrary-URL fetching, 50 MB pre-auth body parsing, unmetered AI calls, an unbounded cache), a login limiter that lets a stranger lock out the whole team, concurrent-operator data loss in the incident blob, an air-quality adapter that never renders the official monitors, and a GPU context-loss recovery path that throws inside every layer's cleanup. None is architecturally difficult.

The MVP recommendation

Feature scope is not what stands between this codebase and a first customer; hardening is. The recommended MVP is the current production-ready core — globe, hazard layers, asset tracking, Property Watch and dashboard, the wildfire risk report and fuel analyzer, the full crisis workspace with sharing and after-action reporting, authentication and admin

— shipped after closing the security gate in the bug report, provisioning the free keys the product silently depends on, deciding the vessel-tracking budget, documenting the two required environment variables, and removing dead code. Fleet tracking, the intelligence feed and the location-details panel ship with stated caveats; the hidden 3D imagery layer, the placeholder crisis tabs and the share-page-only intel pins are deferred.

How to read this paper

Chapter 2 describes the product and its users. Chapter 3 covers the architecture, data tier, identity model, configuration and deployment. Chapters 4 and 5 are the two chapters the product owner asked for by name: threat monitoring, domain by domain, and crisis response, end to end. Chapter 6 is the data-source register. Chapter 7 gives the maturity assessment and the MVP feature cut, and Chapter 8 the platform's known limitations. The companion documents are the *Bug and Vulnerability Report* and the *Product Roadmap Analysis* covering V1 through V4.

What GSOC Monitor Is

GSOC Monitor is a browser-based common operating picture for a Global Security Operations Center. It puts every hazard feed a duty officer needs on one 3D globe, watches a fixed portfolio of properties and vessels against those feeds, and — when something happens — turns into an incident-command workspace that the same team uses to run the response and brief stakeholders.

Purpose

The product answers three questions continuously:

1. **What is happening in the world right now?** Weather warnings, wildfire, smoke and air quality, earthquakes, tropical cyclones, lightning, river flooding, power outages, aircraft, ships, satellites and open-source news are rendered live on a single globe with consistent click-to-detail behaviour.
2. **Does any of it touch our assets?** A hardcoded portfolio of 13 property groups (31 locations — national-park lodging concessions, corporate offices, an airport hangar, a resort, a railway and tour operators) and a seven-ship cruise fleet is scanned against the feeds at fixed distance rings, scored 0–100 per property, and summarised in a status dashboard with an optional AI-written duty-officer briefing.
3. **How do we run the response?** A full-screen crisis workspace holds the incident record — situation report, ICS organisation chart, action log, checklists, intake questionnaire, drawn map layers — synchronises it across every operator in real time, publishes password-protected share links for executives and partners, serves the Incident Action Plan, and produces an after-action report when the incident stands down.

Who uses it

- **Duty officers and analysts** keep the globe on wall displays and desktops, toggle layers, open detail panels, run the Property Watch scan and read the dashboard. Four cinematic screensaver tours keep an unattended display informative.
- **Incident commanders and section chiefs** work inside the crisis workspace: they log actions, assign ICS roles, tick checklist items and draw incident geometry. Several operators can work the same incident at once.
- **Stakeholders outside the team** — executives, property managers, partner agencies — receive a share link and see a phone-friendly read-only situation page with live hazard cards, the map, the IAP and the checklists, without an account.
- **The administrator** manages team accounts, the signup code and the IAP document library.

Operating model

The application is a single deployment serving one team. Everyone signs in with an email, a password and a shared invite code; the first account becomes the administrator. There is no multi-tenant model: the monitored properties, the fleet roster, the tracked aircraft, the checklist template and the AI briefing's context are compiled into the code for the current operator, a hospitality and cruise-line portfolio.

It runs as one Node process on one Heroku dyno with a Postgres database. Public data feeds are pulled from roughly forty providers, nearly all free; most map imagery and several feature services are fetched straight from the operator's browser to keep load off the server, and the server proxies, caches and aggregates the rest. Seven background collectors keep aircraft, vessel, lightning, wind, river, outage and intelligence state current whether or not anyone is watching.

Two modes of operation

Monitoring mode is the default globe: layers, panels, the dashboard and the widgets. It is read-mostly, and every feed degrades independently — a dead upstream leaves a visible status line, never a blank screen.

Crisis mode is entered with the red CRISIS button. The globe docks into a map window inside the incident workspace, and the incident record becomes the shared document of the response. When the incident closes, a three-step stand-down publishes the final snapshot to every share link, revokes the links, releases open ICS assignments and archives the record; a Reopen path exists if the situation changes.

What this paper covers

The following chapters describe the platform from the code as it stands: the architecture and its resilience design, the threat-monitoring capabilities by hazard domain, the crisis-response workflow end to end, the full data-source register, a maturity assessment with the recommended MVP feature set, and the platform's known limitations. A companion bug and vulnerability report and a version roadmap accompany this paper.

Platform Architecture

GSOC Monitor is a three-tier web application: a browser-side 3D globe built on CesiumJS, a single Node/Express process that proxies, caches and aggregates public data feeds, and a Heroku Postgres database that holds the team's own records. The whole system runs on one Heroku "basic" dyno and is designed so that a database outage, a dead upstream feed, or a lost GPU context degrades one feature rather than the whole console.

System overview

The platform has two data paths, and knowing which path a feed uses explains most of its behaviour.

- **Browser-direct feeds.** Basemap tiles (Esri, NASA GIBS, OpenTopoMap), Cesium ion terrain and buildings, RainViewer radar tiles, NASA FIRMS hotspots, NIFC/WFIGS fire incidents, NHC hurricane services, Blitzortung live lightning, BigDataCloud reverse geocoding, Cloudinary uploads, the LANDFIRE fuel ImageServer, the WPC precipitation MapServer, and the county-polygon file are fetched straight from the operator's browser. These providers are CORS-enabled and free, and keeping tile and feature traffic off the single dyno is a deliberate design choice. The cost is that corporate network filtering can blank a layer while the server is healthy, so every layer shows an explicit status line.
- **Same-origin /api proxies.** Everything else goes through the Express server: NWS alerts (as a fallback), USGS earthquakes, the RainViewer manifest, HMS smoke, AirNow/PurpleAir air quality, the NWCG fire outlook, NWPS river gauges, lightning history, ADS-B flights, satellites, ships, wind, outages, JTWC, geocoding, news, GDELT, the OSINT intel feed, and all of the team's own data (auth, incidents, share links, watchlist, IAP documents, briefing).

The server exists for four reasons: to attach identifying User-Agent strings that provider policies require (NWS, Nominatim, Celestrak), to hold API keys that must not reach the browser, to run background collectors that keep state while no client is connected, and to cache aggressively so a room full of wall displays produces one upstream request instead of twenty.

The globe client

The client is a React 18 + Vite 5 single-page application with CesiumJS 1.121, Zustand for state, Tailwind for styling and react-rnd for panel windows. One `Cesium.Viewer` is created for the lifetime of the session and never rebuilt for navigation; when the crisis workspace opens, the globe is pinned into the workspace's map window by resizing its container rather than re-mounting it, so camera position, imagery and every loaded layer survive.

Basemaps. Five map types are available from the sidebar: Dark and Light (Esri gray canvas with a separate label overlay), Dark Satellite (Esri World Imagery, darkened in-shader, with boundary labels), Topographic (OpenTopoMap), and Earth — NASA's daily MODIS true-colour mosaic served by GIBS. The Earth basemap has a floating date navigator that steps one day at a time back to February 2000 (Terra) or July 2002 (Aqua), toggles the AM and PM passes, and rolls forward automatically across UTC midnight so an unattended display keeps showing the latest complete mosaic. Label overlays fade out below roughly 80 km of camera height so town names never sit on top of terrain at close range, and data layers are inserted beneath the labels so city names stay legible over weather.

Rendering environment. Cesium's stock chrome is disabled entirely. The globe renders in `requestRenderMode` over a transparent WebGL canvas with a deterministic 2D starfield behind it, real sun lighting for the day/night terminator, fog and atmosphere. An optional 3D Buildings & Terrain toggle streams Cesium World Terrain and the global OSM Buildings tileset with GPU memory capped at 192 MB plus 128 MB overflow. A Google Photorealistic 3D Tiles layer is wired but hidden from the UI and forced off on every load because the API is metered.

Wall-display hardening. Because the console runs unattended, a lost WebGL context triggers a full in-place viewer rebuild (up to four attempts), and if no working viewer appears within seven seconds the page reloads itself, bounded to three reloads per two minutes. A two-second watchdog also reloads when the GPU reports a lost context or when frames stall for more than eight seconds during a continuous screensaver render. A render-quality tier store and GPU auto-tune exist in the code but are no longer wired to any control — every device gets the highest-quality renderer.

Layer pattern. Every data layer follows the same recipe: a React component that owns one Cesium `CustomDataSource`, a Zustand status store the sidebar reads for its one-line status text, a visibility-aware poller that skips ticks while the tab is hidden, and panel metadata stamped onto each entity. A single global click handler drill-picks up to 16 overlapping features and either opens a docked panel or a "N features here" chooser.

Panels and chrome. Detail panels are draggable, resizable windows that snap into four dock zones; each panel body has its own error boundary so a malformed payload cannot blank the app. Below 768 px the windows become a swipeable bottom sheet. The top bar carries the live clock (local and UTC), a tracked-object counter with sparkline, camera reset, fullscreen, screenshot, the measure tool, the Status dashboard button, the Crisis button with its active-incident badge, the widget launcher (Property Watch, Breaking News, Intel Feed), search, and an information popover that documents the provider and trust basis of every active layer. The sidebar groups layer toggles into Base Map, Hazards & Alerts, Weather, Fire & Smoke, Tracking and Locations sections whose collapsed state persists.

Search and measurement. Search accepts a raw `lat, lon` pair or a place name (server-proxied Nominatim, five results, 24-hour cache). The measure tool computes geodesic distance, spherical polygon area and ellipsoid-corrected radius circles with keyboard shortcuts; terrain height is ignored.

Screensaver and kiosk modes. Four cinematic tours are built in — Global (curated geopolitical landmarks plus live M5+ earthquakes and NWS alerts), National Parks (seven stops with trivia and glowing park boundaries), ISS (SGP4-propagated follow camera), and Pins (an orbit of every monitored property and fleet ship with live hazard cards, a sonar ring for ships and a severity meter in the clock card) — plus a Hover mode that orbits any clicked point. Optional voice announcements use the Web Speech API. There is no idle-timeout auto-start; a mode must be chosen manually and is deliberately not persisted across reloads.

Persisted client state. Layer and basemap selections, sidebar collapse state, crisis dock width, voice preference and the GPU reload budget persist in the browser. Open panels, screensaver state, measure geometry and the Earth basemap date do not.

The server

The server is one Express 4 process. It mounts 29 routers, serves the Vite build as static files, and starts seven background jobs at boot.

Router families.

Family	Routers	Auth
Identity and admin	<code>/api/auth</code> , <code>/api/admin</code>	login/signup public; admin routes admin-only
Team records	<code>/api/incidents</code> , <code>/api/iap</code> , <code>/api/watchlist</code>	signed-in user (IAP upload/delete admin-only)
Crisis sharing	<code>/api/crisis</code>	publish/renew/revoke require sign-in; viewer reads are public behind a per-link password
OSINT	<code>/api/intel</code> , <code>/api/news</code> , <code>/api/news-map</code>	public

Family	Routers	Auth
Hazard feeds	<code>/api/alerts</code> , <code>/api/earthquakes</code> , <code>/api/radar</code> , <code>/api/smoke</code> , <code>/api/aqi</code> , <code>/api/fire-outlook</code> , <code>/api/rivers</code> , <code>/api/lightning</code>	public
Movement and geography	<code>/api/flights</code> , <code>/api/ships</code> , <code>/api/satellites</code> , <code>/api/wind</code> , <code>/api/outages</code> , <code>/api/jtwc-invests</code> , <code>/api/geocode</code> , <code>/api/county</code> , <code>/api/park</code> , <code>/api/park-news</code> , <code>/api/directions</code> , <code>/api/drive</code>	public
Decision support	<code>/api/briefing</code>	public

Two of those routers, `/api/directions` and `/api/drive` , are dead code: the client now calls Overpass and OSRM directly from the browser.

The shared cache. A single in-memory TTL cache backs almost every proxy route. It holds up to 2,000 live entries with insertion-order eviction, keeps a separate 500-entry "last good" map so a failed upstream refresh serves stale data instead of an error, and coalesces concurrent misses into one upstream call. The code is explicit that this is a single-dyno design and should be swapped for Redis before running more than one instance.

Background collectors. Seven jobs run regardless of whether any client is connected:

Job	What it does
ADS-B tracker	Polls adsb.fi for a fixed roster of ten aircraft (10 s while a client is watching, else 60 s), maintains trails and takeoff/landing events, persists to Postgres at most every 30 s
AIS stream	Holds a WebSocket to aisstream.io filtered to the seven fleet vessels, with a by-IMO poller (CruiseMapper scrape by default, VesselFinder or MyShipTracking if a paid key is set) every 120 minutes
Lightning collector	Holds a WebSocket to Blitzortung relays, keeps one strike in six in a 3-million-slot ring buffer spanning 24 hours, and writes gzip'd five-minute chunks to Postgres
Rivers warmer	Refreshes the ~13 MB NWPS national gauge list every 15 minutes so <code>/api/rivers</code> never blocks on the ~50-second upstream pull; pauses after 30 idle minutes
Wind grid	Refreshes a 5-degree global GFS wind grid from Open-Meteo every 30 minutes and upserts it to Postgres; a baked-in grid guarantees a floor
Outage aggregator	Rebuilds an 18-source, multi-state outage picture every 4 minutes; pauses after 15 idle minutes
Intel ingest	Every 90 seconds fans out over seven built-in and every team-added watchlist source, normalises items, geocodes them under a budget, and keeps an 800-item / 36-hour rolling buffer

Live sync. Two Server-Sent Events streams push changes: `/api/incidents/events` to signed-in editors and `/api/crisis/share/:token/events` to share-link viewers, each with a 25-second keepalive. Gzip compression is disabled for these streams because it previously held events in the zlib buffer. Client sets are in-process memory, so live sync assumes one dyno.

Boot and resilience. The process binds its port immediately, then runs schema migrations in the background with exponential backoff (six attempts). If Postgres is unreachable the globe and every non-database route keep working; only auth, incidents, crisis and watchlist error. Unhandled promise rejections are logged rather than fatal, async handlers on the main routers are wrapped to return a fast 503, and a final middleware turns any thrown error into JSON. Static assets under `/assets/` are content-hashed and cached for a year; `index.html` is never cached, and a request for a missing hashed chunk returns 404 rather than HTML so a stale tab cannot blank itself after a deploy.

Data tier

The database schema is created by one idempotent migration script that re-runs on every boot. There is no migrations version table and no foreign keys.

Table	Holds
<code>users</code>	id, email (unique), name, bcrypt password hash, role (<code>member</code> or <code>admin</code>), <code>created_at</code>
<code>settings</code>	key/value pairs: the current signup code and the IAP-seeded flag
<code>incidents</code>	one JSONB document per crisis incident (the whole record, images as Cloudinary URLs), plus timestamps
<code>share_links</code>	token, incident id, frozen snapshot JSONB, active flag, double-SHA-256 password hash, 72-hour <code>expires_at</code> , label, <code>revoked_at</code>
<code>share_access_log</code>	one row per viewer open: token, time, IP, user agent (indexed by token and time)
<code>watchlist_sources</code>	team-shared OSINT sources: kind, URL, label, JSONB config, active flag, added-by user
<code>lightning_chunks</code>	gzip'd five-minute strike chunks forming a 24-hour history that survives deploys
<code>snapshots</code>	small key/value JSONB blobs: the latest wind grid and the flights tracker state
<code>iap_documents</code>	Incident Action Plan PDFs as BYTEA, one per incident type plus a general default (15 MB cap)

Persistence tiers differ by feed. Wind serves live → Postgres snapshot → baked-in fallback. Flights and lightning restore from Postgres on boot. The ship tracker persists only to a JSON file on the dyno's filesystem, which Heroku wipes on every deploy. The intel buffer and every cache are memory-only and start empty after a restart.

Identity and access

Authentication is self-contained. Signing up requires a shared invite code (rotatable from the admin panel or pinned by an environment variable); the very first account becomes the admin and every later one is a member. Login issues a JSON Web Token carrying id, email, name and role, valid for 30 days, in an `httpOnly`, `sameSite=lax` cookie; passwords are hashed with bcrypt at cost 12. Tokens are verified by signature only — the user table is not consulted on later requests. An in-memory counter blocks login and signup after ten failures from one IP within ten minutes.

The admin panel lists and deletes users, rotates the signup code, and manages the IAP PDF library. There is no role promotion, password reset, email verification or multi-factor authentication.

Every live data-layer route and the public OSINT feed are unauthenticated. Share links are the only public door into team data, gated by a per-link 10-character password (the browser sends a SHA-256 of it as a query-string key), a 72-hour renewable expiry and a revocation stamp, with each open written to the access log.

Configuration and deployment

All configuration is by environment variable. Only two are required in practice, and neither is documented in the repository's example file or Heroku manifest.

Variable	Purpose	Required	Documented
<code>DATABASE_URL</code>	Postgres connection string	yes	no
<code>JWT_SECRET</code>	signs session cookies; every auth call fails without it	yes	no

Variable	Purpose	Required	Documented
PORT , NODE_ENV	set by Heroku; NODE_ENV=production enables the cookie's secure flag	platform	partly
NWS_USER_AGENT	identifying User-Agent for NWS, Nominatim, CelesTrak, adsb.fi	recommended	yes
SIGNUP_CODE	pins the invite code; re-applied on every boot, overriding admin rotations	no	no
AISSTREAM_API_KEY	free aisstream.io key; ships layer is empty without it	no	no
VESSELFINDER_API_KEY , MYSHIPTRACKING_API_KEY	optional paid by-IMO vessel positions	no	no
SHIPS_SCRAPES_CRUISEMAPPER , AIS_MMSI_FILTER , SHIPS_POLL_MINUTES , SHIPS_SNAPSHOT_PATH	ship tracker tuning	no	no
AIRNOW_API_KEY , PURPLEAIR_API_KEY	air-quality providers; layer is empty without them	no	no
NPS_API_KEY	park alerts/news; defaults to the rate-limited DEMO_KEY	no	no
ANTHROPIC_API_KEY , BRIEFING_MODEL	enables the AI duty-officer briefing; rules digest otherwise	no	no
VITE_CESIUM_ION_TOKEN	build-time; terrain and buildings fall back to Cesium's rate-limited demo token	no	no
VITE_GOOGLE_MAPS_KEY	build-time; hidden photorealistic 3D layer	no	no

Deployment is a single Heroku dyno via the Node buildpack: `npm install` , `npm run build` (Vite build, then TypeScript compile), `npm start` . The `app.json` manifest declares the buildpack and one basic web dyno but no Postgres add-on. Continuous integration runs on every push — install, typecheck of both workspaces, Vitest in both workspaces, and a full build — with no Postgres service and no deploy step.

Test coverage. Twenty-two unit-test files exist: five on the server (cache, checklist validator, incident taxonomy, aircraft-info parsing, flight state machine) and seventeen on the client (crisis store and sync canonicalisation, AAR metrics, checklist and intake templates, layer measurement, flight markers, fuel reference, POI classifier, directions client, geodesy, measure math, and the wildfire risk report's level algebra). Nothing exercises an HTTP route end to end, authentication, migrations, or any map layer component.

Extension model

The README's five-step recipe for adding a layer still describes the shape of the code: add a layer id to the client types, add a `CustomDataSource` component under `client/src/layers/` , register a detail component in the panel router, add a proxy route under `server/src/routes/` , and add a sidebar toggle. Two details in the README no longer match the code: zone-based NWS alerts are resolved to county polygons in the browser from a static county file, not server-side, and flights are tracked from a fixed server-side roster rather than refreshed from the camera viewport.

Threat Monitoring

GSOC Monitor watches eleven threat domains and fuses them into property-centric judgments for a duty officer. Each domain is a CesiumJS layer with a sidebar toggle, a status line, click-to-open panels and polling that pauses while the tab is hidden; server-routed feeds share one TTL cache with stale-on-error and request coalescing (`server/src/cache.ts`), while NWS alerts, FIRMS, NIFC, NHC, LANDFIRE and live lightning are fetched directly from the browser.

Severe weather

Four layers: NWS alert polygons, RainViewer radar and infrared satellite, the WPC precipitation forecast raster, and a global wind field.

NWS alerts

"NWS Alerts" (on by default) draws every active NWS watch, warning and advisory as a translucent polygon. Storm-based warnings carry their own geometry; alerts shipped with `geometry:null` are resolved client-side by matching SAME county codes against a static ~3,200-county FIPS GeoJSON from a Plotly GitHub mirror, so zone-only alerts (marine, fire-weather zones) draw nothing. Colour is by hazard type across about fifteen hues, falling back to severity (Extreme, Severe, Moderate, Minor), with the most severe drawn on top; there is no on-map colour key. The panel shows the full CAP text (headline, severity, urgency, certainty, timing, area, description, instructions). The browser fetches `https://api.weather.gov/alerts/active` directly every 60 s (20 s timeout), falls back to `/api/alerts/active` (60 s server cache, `NWS_USER_AGENT` header), and prefers the proxy for 10 minutes before retrying direct.

Precipitation radar and IR satellite

"Precipitation Radar" loads the RainViewer manifest through `/api/radar` (2-minute cache and poll) and stacks one imagery layer per frame; tiles come straight from RainViewer's CDN. Playback runs at 800 ms per frame with a 720 ms crossfade, looping from the observed window (30, 60 or 120 minutes; default 120, twelve ten-minute frames) into nowcast frames, with a bottom timeline scrubber. Controls set Radar, Clouds or Combined mode, four palettes and opacity. Because the CDN ignores its colour-scheme path segment, tiles are recoloured client-side, a documented workaround that still ships a temporary `/api/radar/diag` probe. Combined mode pairs each radar frame with its nearest IR frame within 60 minutes and is `partial`, not the default pending calibration.

WPC precipitation forecast

"Precipitation Forecast (WPC)" renders Weather Prediction Center QPF accumulation polygons as live ArcGIS export tiles clipped to CONUS (max zoom 9) for 24 h, 48 h, 72 h (default) or 5 days, with an opacity slider (default 0.75) and a floating legend of the 18-stop WPC ramp from 0.01 in to 20+ in. There is no app cache and no error latch, so a NOAA outage shows as blank tiles. The product is issued at 06Z and 18Z.

Global wind field

"Wind (GFS)" advects 7,000 particles on the CPU at about 30 fps through a bilinearly interpolated global 10 m wind grid, lifted 3 km and coloured by speed across 0–45 m/s. "Wind Streamlines" integrates the same field 18 steps forward from a camera-adaptive seed grid (at most 1,600 lines) into glowing polylines with surface-tangent arrowheads. The

probe (off by default) adds a cursor-following arrow and a HUD readout (compass, speed in mph, kt or m/s, direction, a HISTORICAL badge when stale); right-click pins a point whose panel shows a 48-hour hourly chart and a 7-day daily outlook.

The grid is 72 × 33 points at 5 degrees (2,376 points, latitude ±80) from Open-Meteo GFS in five batches of 500, refreshed server-side every 30 minutes, upserted into the Postgres `snapshots` table so a redeploy restores real wind, with a committed baked grid as last resort; anything older than 45 minutes is flagged stale. The client refetches every 30 minutes and hydrates from an ~80 KB localStorage copy. Point forecasts (`/api/wind/forecast` , 7-day hourly, 30-minute cache at 0.1-degree rounding) fall back to MET Norway on a 10-minute circuit breaker.

Wildfire, smoke and air quality

Six layers: satellite heat detections, named incidents, the seven-day outlook, smoke plumes, fuel models with a draw-a-zone analyzer, and air quality.

FIRMS VIIRS hotspots

"Wildfires (NASA FIRMS)" draws past-24 h VIIRS 375 m detections from the Esri Living Atlas FeatureServer, fetched directly by the browser. Viewport mode queries the camera rectangle, capped at the 2,500 strongest detections by Fire Radiative Power, re-running on camera stop (800 ms debounce) and every 5 minutes, with flames sized and coloured by FRP tier at 100, 30 and 8 MW. "Near pins" switches to one envelope query per property group filtered to 5, 50, 100 or 200 miles of a pin, the same function that feeds Property Watch and the risk report. The panel grades each detection from FRP and I-4 brightness (intense front at 75 MW, flaming at 15, active at 3, smouldering at 0.8, with brightness-temperature equivalents) and warns that a detection is a heat signature, not a confirmed wildfire. No on-map legend.

Named incidents and perimeters (NIFC/WFIGS)

"Named Fires (NIFC)" fetches interagency incidents directly from the WFIGS ArcGIS services every 5 minutes: wildfires of 5 acres or more under 100 percent containment (largest first, at most 800) plus perimeters for incidents of 100 acres or more (at most 500, simplified to roughly 800 m). Markers are coloured by containment (under 30 percent or unknown, under 70, under 100, contained) with an acreage label; a perimeter failure never blocks the markers. The panel shows containment, acres, personnel, complexity, cause and an InciWeb link. Perimeters are not clickable.

7-day fire potential (NWCG)

"7-Day Fire Potential (NWCG)" shades about 234 Predictive Service Area polygons for the selected day with the official palette, IGNITION and CRITICAL overriding fuel dryness (Very dry, Dry, Normal). The server caches the seven-day payload 2 hours; the client polls every 30 minutes and the day buttons re-render from cache. This is the one hazard layer with a floating legend. CONUS only.

Smoke plumes (NOAA HMS)

"Smoke (NOAA HMS)" draws analyst-delineated plumes from the Hazard Mapping System daily KML, filled by density (Light, Medium, Heavy on top). The server downloads today's UTC file, falling back up to two days, and caches 2 hours; the client polls every 2 hours. The panel gives density, satellite, times and product date. North America only.

Fuel models and the Fuel Analyzer (LANDFIRE)

"Fuel Models (LANDFIRE)" renders the LF2024 Scott and Burgan FBFM40 raster (30 m, CONUS, max zoom 16) as live USGS ImageServer tiles at 0.72 alpha with the official colormap. The 45-class codebook (91–99 nonburnable, 101–204 burnable) rolls up into seven groups, Grass, Grass-Shrub, Shrub, Timber-Understory, Timber-Litter, Slash-Blowdown and Nonburnable, shown as swatches in the floating FBFM40 legend. The Fuel Analyzer lets the operator

draw a circle (150 m to 250 km) or a polygon; the ring is POSTed from the browser to the ImageServer `computeHistograms` endpoint and decoded with an affine bin-to-value mapping into exact per-class pixel counts. The panel reports area (900 m² per cell), burnable percentage, per-class and per-group shares, and an app-defined 0–100 fire-behaviour potential score (Low under 20, Moderate under 40, High under 60, Very High under 78, else Extreme) with spread and flame-length categories and up to three drivers; tapping a model pins the published RMRS-GTR-153 reference card. The score tables are heuristics at standard fire weather (10 mph wind, 5 percent 1-h fuel moisture) with no live adjustment.

Air quality (AirNow + PurpleAir)

"Air Quality (AirNow + PurpleAir)" draws CONUS stations from `/api/aqi`: AirNow reference monitors as numbered badges in EPA category colours, PurpleAir community sensors as dots. AirNow supplies hourly observations of six pollutants over a 3-hour window inside a CONUS bounding box, the highest-AQI pollutant dominant. PurpleAir supplies outdoor sensors reporting within the hour with confidence 70 or better, corrected with the EPA Barkjohn 2021 formula $PM_{2.5} = 0.524 \times cf1 - 0.0862 \times RH + 5.75$ (RH defaults to 35), converted with the 2024 breakpoints and stride-sampled to at most 6,000 sensors. The merged set is cached and polled every 15 minutes; per-source toggles filter without a refetch. `AIRNOW_API_KEY` and `PURPLEAIR_API_KEY` are **KEY REQUIRED** and undocumented; without them the layer shows only a "set key" hint.

Seismic

"Earthquakes (USGS)" (on by default) draws each event as ripple rings sized and coloured by tier (M6+, M4.5+, M2.5+, minor). The server proxies the USGS summary feeds at `/api/earthquakes?magnitude=&period=` (significant, 4.5, 2.5, 1.0, all; hour, day, week) with a 60 s cache. Sidebar buttons M1.0+, M2.5+, M4.5+ and Significant set the magnitude (default 2.5, persisted). The period exists in state and scales the poll (60 s hourly, 5 min daily, 15 min weekly) but has no UI, so operators are locked to the past day; that filter is `partial`. The panel shows magnitude, place, time, depth, felt reports and tsunami flag; the PAGER alert level is not displayed.

Tropical and lightning

Tropical cyclones

"Hurricanes (NHC + JTWC)" (on by default) polls every 5 minutes and draws, per storm, the forecast error cone, the observed track in solid grey, the forecast track in dashed amber, forecast dots tinted by predicted Saffir-Simpson category, and a spiral eye glyph rotating once every 5 s. Hover tooltips give category, advisory time and winds; the panel adds pressure, basin and an NHC link, classified on the 64/83/96/113/137 kt thresholds. NHC and CPHC Tropical Weather Outlook "Potential Development Region" polygons are coloured by 7-day risk (Low, Medium, High) with 48-hour and 7-day odds in the panel. Invests in other basins, such as "96W", are parsed server-side from the JTWC ABPW10 and ABIO10 text bulletins (`/api/jtwc-invests`, 1-hour cache, empty set rather than an error on failure) and drawn as a roughly 100 NM ellipse. NHC and GTWO data come directly from Esri and NOAA ArcGIS with no server cache; wind radii, advisory text and watches/warnings are not rendered.

Lightning

"Lightning (Blitzortung)" opens a WebSocket from the browser to a Blitzortung relay (ws1, ws7 or ws8, rotating), subscribes with `{a:111}` and LZW-inflates each frame. Each strike drops a crosshair that ages through four colours and expires at 10 minutes, capped at 2,500. Window buttons for 1 h, 6 h, 12 h and 24 h enable the history layer, which polls `/api/lightning?minutes=` every 30 s and renders older strikes coloured by age; a legend keys both ramps.

The server runs its own collector on the same relays: it keeps every sixth strike in ring buffers of 3,000,000 slots (about 48 MB), writes gzip chunks to the Postgres `lightning_chunks` table every 5 minutes, deletes chunks older than 25 hours, and restores 24 hours at boot. Queries clamp to 1–1,440 minutes, thin to at most 20,000 points, and accept a lat/lon/radius filter (up to 500 mi) applied before the cap so local storms arrive unthinned; `/api/lightning/debug` reports buffer fill. Because the buffer is a one-in-six sample, "N strikes in the last 6h" is a sample count, and live strikes are not clickable.

Flood

"Rivers & Floods (NWPS)" renders the NOAA National Water Prediction Service gauge list, about 12,700 points, coloured by observed flood tier (Major, Moderate, Minor, Action, Normal, Low, unclassified). Filter chips All, Action+, Minor+, Mod+ and Major and a "Highlight forecast-to-flood" toggle re-render without refetching. The server refreshes the ~13 MB bulk list every 15 minutes in the background (never on the request path), drops out-of-service gauges, and answers `warming:true` on cold start so the client re-polls after 12 s, otherwise every 15 minutes. Clicking a gauge fetches `/api/rivers/:lid` (10-minute cache): stage or flow with a Rising/Falling/Steady trend, a threshold bar from action through major with current and forecast-crest markers, a 14-day observed (160 points) plus forecast (80 points) hydrograph, up to six NWS impact statements, crests and a USGS link. There is no alerting when a gauge crosses a tier.

Infrastructure: power outages

"Power Outages (Multi-State)" draws each outage from `/api/outages` as a bolt-in-ring marker, red for Unplanned, amber for Planned, grey for unknown, labelled "Utility · 1.2k" inside 1,500 km. The panel shows customers affected, start time, estimated restoration (or a red overdue age), area and cause.

The server normalises 18 sources. Five ArcGIS FeatureServers: Cal OES (CA), Salt River Project (AZ), SSVEC (AZ), Minnesota Power / SWLP (MN) and Riverside Public Utilities (CA). Nine KUBRA Storm Center instances, walked by quadkey with a 150-tile / 25 s cap per utility: Oncor (TX), Georgia Power (GA), JEA (FL), Colorado Springs Utilities (CO), Cobb EMC (GA), LG&E / KU (KY), Evergy (KS), Versant Power (ME) and Appalachian Power (VA). Four NISC co-op tenants: Sawnee EMC (GA), SLEMCO (LA), Price Electric (WI) and Cloverland Electric (MI). Those carry 14 state labels; the panel footer quotes roughly 19 states because territories cross state lines. The aggregate is rebuilt every 4 minutes with each source caught independently, paused after 15 minutes without requests; the client polls every 5 minutes. Coverage is far from national, the KUBRA GUIDs are hard-coded and may rotate, and outages are points with no extents. `partial`

Aviation

"Flights (ADS-B)" tracks ten tails anywhere on the globe, polled by registration from `adsb.fi` on the server: company aircraft N10AZ, N14NA and N154LA; NOAA hurricane hunters N42RF "Kermit" and N43RF "Miss Piggy" (WP-3D) and N49RF "Gonzo" (Gulfstream IV-SP); and 10 Tanker DC-10s N17085, N522AX, N603AX and N612AX (Tankers 910, 911, 912, 914). Company tails poll every 10 s while a client has hit the route in the last 5 minutes (60 s otherwise) and are held at last-known position forever; the special rosters poll every 60 s, appear while flying and for 2 hours after the last fix, then clear until the next mission. USAF WC-130J hunters cannot be tracked because they fly military hexes under mission callsigns.

Markers are silhouettes rotated to ground track with a nametag, tinted on the tar1090 altitude rainbow (orange at 2,000 ft, green at 10,000 ft, magenta at 40,000 ft); grounded aircraft turn grey, and after 180 s without a fix an aircraft is clamped to the ground and dimmed. Trails are altitude-coloured polylines with chevrons every 30 km and a break at gaps over 15 minutes, kept 24 h for company tails and 2 h for specials (2,500 points, 700 on the wire). Ground-state

transitions become takeoff or landing events after a 2-minute confirmation hold; 200 are retained and the four most recent appear in the sidebar resolved to a nearby city. The panel shows a Planespotters photo, make, model and operator from adsbdb (refreshed every 7 days), altitude, speed, heading, vertical rate and squawk. State is upserted to the Postgres `snapshots` key `flights:v1` at most every 30 s and restored at boot; the client polls every 10 s. The roster is hard-coded with no UI to add tails.

Maritime

"Ships (AIS)" tracks the seven Windstar Cruises vessels, Star Breeze, Star Legend, Star Pride, Star Seeker, Wind Spirit, Wind Star and Wind Surf, by MMSI and IMO. Each is a sonar-contact marker that fades after 20 minutes and again after 2 hours without a report. "Show past & future paths" adds the 72-hour trail (at most 400 points) and a dashed 6-hour dead-reckoning projection along course above 0.5 kt. The client polls `/api/ships` every 30 s.

Positions merge freshest-wins from two paths. A server WebSocket to AISStream (free `AISSTREAM_API_KEY`, filtered to the fleet MMSIs) receives reports when a ship is within a community receiver's range, which in practice has not happened. The by-IMO poller, every `SHIPS_POLL_MINUTES` (default 120, minimum 15), therefore does the real work: a CruiseMapper page scrape behind Cloudflare by default, or Vesselfinder (about 2,500 credits a month at the default cadence) or MyShipTracking when a key is set, both **PAID**. Positions persist to a dyno-local JSON file wiped on every deploy; ships stay at last-known position indefinitely. The panel shows a rotating wireframe class card and the AIS voyage fields; `/api/ships/debug` reports socket state and scrape blocks. "Copy fleet snapshot" renders a 1,520 px PNG (coastline map, 72 h tracks, callout chips geocoded via BigDataCloud, fleet table), copied to the clipboard and downloaded as `windstar-fleet-YYYY-MM-DD.png`. **partial**

Space

"Satellites (CelesTrak)" loads two-line element sets for one of five groups, Space Stations, Brightest, GPS, Weather or Starlink, from `/api/satellites?group=` (2-hour server cache) and propagates every satellite in the browser with `satellite.js` SGP4 on a tick of 250 ms (up to 200 satellites), 800 ms (up to 800) or 1,500 ms. A cap of 2,500 truncates Starlink; and the ISS keeps a gold glyph. Groups of 500 or fewer drag a fading trail, and the selected satellite gets a full-revolution orbit ring of 160 SGP4 samples recomputed every 8 s. The panel re-propagates every second with position and orbital elements. "Track the ISS" selects the stations group and flies to NORAD 25544 at 5,000 km with a -35° pitch. TLEs are fetched once per toggle and never refreshed while the layer stays on.

Open-source intelligence

Three server pipelines: a Breaking News RSS aggregator, a GDELT geocoded news map, and a Dataminr-style Intel Feed with a team-shared watchlist.

Breaking News

`/api/news` fetches five hard-coded feeds, BBC News World, The Guardian World, Sky News World, NPR News and Al Jazeera, caches the merged set 5 minutes with a 60 s stale cooldown on failure, and caps at 120 stories. Each story is scored critical, urgent or alert by keyword lists; categorised by regex into conflict, disaster, weather, politics, economy, health or environment; and geolocated by a static gazetteer of about 45 hotspot cities and 50 country centroids, so most headlines get no coordinates. The widget polls every 5 minutes with severity and category chips and a Sources section where users add RSS URLs stored per browser and sent as `?extra=` without public-host validation. A "Park News" mode swaps to NPS alerts for six parks (10-minute cache), and a bottom ticker shows the stream when the panel is closed.

GDELT news map

`/api/news-map` queries the GDELT GKG GeoJSON v1 endpoint with a default security query (protest, evacuation, wildfire, shooting, explosion, flooding, active shooter, lockdown, hazmat, riot, derailment) over 6 hours, at most 250 rows, grouped into one pin per location with article count, average tone, an image and up to six links, cached 10 minutes. Pins are sized by count and red when tone is -5 or lower. The layer is forced off in the operator console on every store hydration and mounts only on crisis share-link globes; operators see GDELT only through the dashboard's News metric. `partial`

Intel Feed

The intel engine runs a cycle every 90 s over seven built-in sources plus every active row of the Postgres `watchlist_sources` table: a "National Park Alerts" Google News search and two regional searches pinned to the Yellowstone and Grand Canyon regions, a Bluesky "wildfire evacuation" search, the PulsePoint San Francisco Fire/EMS CAD (a reverse-engineered endpoint), the California CHP dispatch XML, and the City of Chicago crime dataset on Socrata. Every item is normalised to one shape with a keyword-regex severity of info, watch or urgent and a category among scanner, crime, crash, fire, weather, news, social and other. Items with a place but no coordinates are geocoded through OSM Nominatim under a budget of 16 lookups per cycle spaced 1.1 s apart, cached 30 days, with misses blacklisted. The buffer is capped at 800 items or 36 hours, served publicly at `/api/intel` without authentication, and lost on restart.

The widget polls every 90 s and lists up to 150 rows; clicking flies to a geolocated item or opens its URL. Any signed-in member can add a team-wide watchlist source from seven presets: RSS URL, Google News topic, Bluesky account, Bluesky keyword, PulsePoint agency id, Socrata domain plus dataset id, or CHP dispatch, with an optional place pin the server geocodes. Rows are validated per kind, capped at 100, and can be paused or deleted. On-globe intel pins are, like GDELT, forced off in the operator console and render only on share pages (`partial`). All kinds share the 90 s cadence, there is no push alerting, and severity is keyword matching rather than NLP.

Decision support

Four surfaces turn the feeds into judgments about the monitored estate: 13 property groups holding 31 locations (national parks, offices, an airport, a resort and a railway), hard-coded in `client/src/layers/locations/locations.ts`, plus the seven-ship fleet.

Property Watch

The "Watch" widget scans at 25, 50 or 100 mi (default 25). It fetches in parallel the FIRMS hotspots near pins (13 envelopes), NWS active alerts with the county GeoJSON, and USGS M2.5+ earthquakes from the past week, then for each of the 31 locations lists hotspots within the radius nearest-first with FRP, quakes nearest-first, and alerts whose polygon contains the point. Only properties with a hazard are returned, ranked by worst alert severity then nearest physical hazard. The widget polls every 5 minutes and the shared store throttles to one scan per 60 s; the status line names any feed that is down, and the all-clear banner is suppressed when every feed has failed. A per-property pop-out lists alerts, nearby fires and recent quakes, opens the wildfire report, and exports a PNG watch card.

Property Status Dashboard

The TopBar "Status" button opens a full-screen overlay listing every group as a card sorted alert, watch, ok, rescanning every 300 s while open. Status derives from the 100 mi scan: alert when the worst NWS alert covering any location has severity rank 3 or higher (Severe or Extreme) or the nearest FIRMS hotspot is under 25 mi; watch when any alert is in effect, a hotspot is within 100 mi, or an M2.5+ quake from the past week is within 100 mi; otherwise ok. Each card shows alerts, Open-Meteo weather and 7-day rain for the primary location, the nearest AQI station (75 mi), hotspot,

quake, GDELT events (150 mi) and outage (100 mi). A live feed holds up to 120 deduplicated events of seven types: alert, fire, quake, news (up to four per group), outage, storm (tropical systems within 250 mi of a property or 300 mi of a fleet ship, from NHC observed positions plus JTWC invests) and ais ("AIS signal silent for Nh" after 6 hours of fleet silence); the storm and AIS checks are the least hardened parts. The dashboard scans only while open, keeps no history and resets its feed on reload.

The threat score

`threatScore.ts` computes a per-group 0–100 score as the sum of additive contributions, rounded and clamped to 100, with the top three reasons attached:

Signal	Contribution
NWS alerts	base 45 if worst severity rank ≥ 4 (Extreme), 35 if ≥ 3 (Severe), 18 if ≥ 2 (Moderate), else 8; plus $\min(12, (\text{alert count} - 1) \times 4)$
Nearest hotspot	30 if < 10 mi, 22 if < 25 mi, 12 if < 50 mi, else 6 (within the 100 mi scan)
Nearest earthquake	15 if magnitude ≥ 5 and < 50 mi, 8 if magnitude ≥ 4 , else 4
Nearest outage	10 if < 25 mi else 5, plus 5 if customers $> 1,000$
Air quality	8 if AQI > 150 , 4 if AQI > 100
7-day forecast rain	6 if ≥ 3 in, 3 if ≥ 1.5 in
Current wind	6 if ≥ 35 kt, 3 if ≥ 25 kt

Severity ranks are Extreme 4, Severe 3, Moderate 2, Minor 1, Unknown 0. The chip is red at 60 or more, orange at 35, yellow at 15 and green below, hidden at zero, with reasons on hover. The weights are constants in code, described in the source as deliberately simple and explainable rather than calibrated.

The duty-officer briefing

The briefing panel shows a headline, two or three paragraphs and a source tag. The client fuses the scan into a signals payload (per-group level, score, reasons and metrics plus the twelve most recent feed lines) and POSTs it to `/api/briefing`, throttled to once per 9 minutes unless forced. With `ANTHROPIC_API_KEY` set the server calls the Anthropic SDK with model `BRIEFING_MODEL` (default `claude-opus-4-8`), 700 max tokens and a system prompt demanding a headline of 12 words or fewer then 100–180 words grounded strictly in the JSON, treating payload text as data. Without a key, or on any API error, a deterministic rules digest is composed ("All properties nominal" or "N properties elevated - highest: X", listing groups scoring 15 or more). Responses are cached 10 minutes by SHA-1 of the payload minus its timestamp, up to 20 keys; payloads are limited to 40 groups and 30,000 characters. The AI path is

PAID.

Property Wildfire Risk Report

From a property pop-out, "Risk" opens a full-screen "Property Risk Report - Wildfire"; only the wildfire hazard is implemented, so the report is `partial`. Assembly fetches eleven feeds in parallel with independent failure, a down feed becoming an "Unavailable" section excluded from the overall level rather than a silent Low: FIRMS hotspots at 100 mi, NIFC incidents and perimeters, NWS alerts filtered to red flag, fire weather, fire warning, extreme fire, smoke and evacuation and point-tested at the site, county geometry, the NWCG outlook, a LANDFIRE zonal analysis on a 3 mi circle (CONUS only), the 48-hour wind forecast, the 10-day daily forecast, HMS smoke (North America only), 24 hours of Blitzortung strikes within 130 mi, and a WPC QPF identify at the point.

Exposure is measured at four fixed rings: 1 mi "Evacuation / direct impact", 5 mi "Mutual aid - staging - access", 25 mi "Supply - staff commute" and 100 mi "Logistics - alternate lodging". Eight sections are rated low, guarded, elevated, high or critical, and the overall level is the maximum of the available sections. Hotspots: nearest within 1 mi critical, 5 mi high, 25 mi elevated, else guarded, bumped one level if any hotspot within 25 mi exceeds 100 MW. Named fires: an incident under 50 percent contained within 25 mi is high, otherwise elevated, bumped at 1,000 acres or more, with a staleness warning when the record is 24 hours old; all contained at 50 percent or more is guarded. Alerts: red flag, evacuation, fire warning or extreme fire high, fire weather elevated, other guarded. Outlook: CRITICAL today high, IGNITION elevated, dryness 2 or more guarded. Fuel: zonal score 85 or more high, 70 elevated, 40 guarded. Wind: 35 mph now or a 48-hour gust of 50 mph high; 25 mph or a 35 mph gust elevated. Smoke: Heavy overhead elevated, Medium guarded. Lightning: a strike within 5 mi elevated, within 25 mi guarded.

The report carries a BLUF card, five stat cards, a ring exposure table, the eight rated sections each with its own map, rainfall and 10-day forecast strips, an eleven-entry sources list and a "Not yet factored" list naming relative humidity, fuel moisture and terrain slope. Seven map figures are rendered off-DOM; a figure that fails to load is omitted. Output is browser print only via `window.print()`; there is no server-rendered PDF, persisted report or share link. Thresholds are mirrored by hand in `docs/RISK-REPORT-MATRIX.md`.

Coverage and freshness at a glance

Domain	Source(s)	Cadence	Geographic coverage	Maturity
NWS alerts	api.weather.gov direct; /api/alerts/active fallback (60 s cache)	60 s	US; county polygons for SAME-coded alerts	production
Radar / IR	RainViewer via /api/radar (2 min cache); tiles direct	2 min; 10-min frames; 30–120 min window plus nowcast	Global mosaic	production; Combined partial
Precipitation forecast	NOAA WPC QPF, live tiles	Live; issued 06Z and 18Z	CONUS	production
Wind field	Open-Meteo GFS via /api/wind ; Postgres snapshot; baked fallback; MET Norway probe fallback	30 min; stale after 45 min	Global 5° grid, ±80°	production
FIRMS hotspots	Esri Living Atlas VIIRS, browser-direct	5 min plus camera stop; 24 h	Global; top 2,500 in view or 5–200 mi of pins	production
Named fires	NIFC/WFIGS ArcGIS, browser-direct	5 min	US; incidents ≥ 5 ac, perimeters ≥ 100 ac	production
7-day fire potential	NWCG via /api/fire-outlook (2 h cache)	30 min	CONUS, ~234 PSAs	production
Smoke	NOAA HMS via /api/smoke (2 h cache)	2 h; up to 2 days old	North America	production
Fuel models	USGS LANDFIRE LF2024 FBFM40, browser-direct	On demand	CONUS, 30 m	production
Air quality	AirNow + PurpleAir via /api/aqi (15 min cache)	15 min	CONUS box; ≤ 6,000 PurpleAir sensors	production; keys required
Earthquakes	USGS via /api/earthquakes (60 s cache)	5 min (day feed)	Global	production; period filter partial

Domain	Source(s)	Cadence	Geographic coverage	Maturity
Tropical	NHC and GTWO ArcGIS direct; JTWC via <code>/api/jtwc-invests</code> (1 h cache)	5 min	NHC/CPHC basins; JTWC W Pacific, Indian Ocean, S Hemisphere	production; no server cache
Lightning	Blitzortung in-browser; server collector, Postgres 24 h	Live; history 30 s, 1–24 h	Global; 1-in-6 sample	production
River gauges	NOAA NWPS via <code>/api/rivers</code> (15 min background); detail 10 min cache	15 min	US, ~12.7k gauges	production
Power outages	18 sources (5 ArcGIS, 9 KUBRA, 4 NISC) via <code>/api/outages</code>	4 min server; 5 min client	14 state labels (~19 served); points only	partial
Aircraft	adsb.fi by registration; adsbdb and Planespotters; Postgres snapshot	10 s company / 60 s specials	Global; 10 hard-coded tails	production
Ships	AISSStream; CruiseMapper scrape or VesselFinder / MyShipTracking (paid)	Stream plus 120 min poll; 30 s client	Global; 7 Windstar vessels	partial
Satellites	CelesTrak via <code>/api/satellites</code> (2 h cache); SGP4 in browser	Once per toggle; 250–1,500 ms tick	Global; 5 groups, ≤ 2,500	production
Breaking News	5 RSS feeds plus custom via <code>/api/news</code> (5 min cache); NPS park news	5 min / 10 min	Global; gazetteer geolocation only	production
GDELT news map	GDELT GKG v1 via <code>/api/news-map</code> (10 min cache)	10 min; 6 h window	Global	partial (share-page pins)
Intel Feed	7 built-in sources plus watchlist (≤ 100) via <code>/api/intel</code>	90 s; 800 items / 36 h	Source-dependent; ephemeral	production; pins partial

Crisis Response

Overview

The crisis workspace is the platform's incident-management mode. The red CRISIS button in the top bar opens a full-screen overlay that replaces the globe with an incident dashboard and, for any open incident, a tabbed workspace with a live-map dock. When any incident is Active the button pulses with a count badge and the browser tab title becomes `CRISIS (N) · GSOC Monitor`.

It serves three audiences: **operators** (any signed-in member — incidents are team-wide with no ownership), **administrators** (who alone delete archived incidents and manage the Incident Action Plan library) and **stakeholders**, who open a password-protected share link on a phone with no account.

Lifecycle

An incident carries one of four statuses in lifecycle order: **Monitoring** (sky, list rank 1) → **Active** (red, rank 0 — sorts first, pulses, counts toward the CRISIS badge) → **Recovery** (amber, rank 2) → **Closed** (green, rank 3). A new incident starts Active. The operator sets any status from a chip row; ordering is not enforced, and every change writes a `status-change` audit entry with `{from, to}`.

Stand Down concludes an incident through three steps, each shown with ✓/✗ status in a modal that previews the plan and takes an optional reason:

1. Publish the final Closed snapshot to every active share link and wait, so viewers see the conclusion rather than a stale ACTIVE page.
2. Revoke every active share link. Failures are non-blocking; an unrevoked link keeps serving the closed snapshot.
3. Archive: stamp `archivedAt`, force Closed, record who and why, end every open ICS assignment, and write a `stood-down` entry with the reason and the number of released assignments.

An archived incident is a frozen record: content edits are no-ops and the UI is disabled, except for log viewing, share-link revocation (a leaked link) and after-action content. **Reopen** clears the archive stamps, sets Monitoring — the operator escalates — writes a `reopened` entry and republishes to any surviving link; the stand-down history stays in the log.

The incident record

Every incident is one self-contained JSON document, minted client-side with a `c-` + UUID id so multiple tabs never collide, and stored as a JSONB row in Postgres. The server validates only id, type and status on write; the rest is opaque, bounded by a 5 MB body limit.

Group	Fields
Situation report	Name; start and end (datetime-local, no timezone); location; type; status; executive summary; ICS complexity type (Type 5 → Type 1, stored and audited but its selector is removed from the UI)
Location, property, vessels	<code>locationGroupId</code> — the primary property group or the pseudo-group <code>windstar-ships</code> ; <code>extraLocationGroups</code> — extra property-pin groups for the share map; <code>shipMmsis</code> — fleet vessels by MMSI only (positions are always read live)
Org chart	<code>roles[]</code> — ICS nodes (title, abbreviation, parent, colour, command-staff and support flags, <code>builtin</code> lock on the 21 NIMS defaults); <code>personnel[]</code> — per-incident pool with contact details; <code>assignments[]</code> — who held which role, with start/end stamps

Group	Fields
Log	<code>actionLog[]</code> — newest-first entries with timestamp, description, <code>action / event / info</code> type, actor, optional Cloudinary image, and for system entries a <code>system</code> kind plus <code>meta</code> map
Checklists	Sparse map of item id → <code>{checked, at, by}</code> , server-stamped
Intake	Sparse map of question id → answer text
Draw layers	<code>drawLayers[]</code> — name, category type, geometry (polygon / line / point), directional flag, colour, visibility, vertices, thumbnail; <code>liveLayers[]</code> — live feeds prescribed for the share map
Share links	<code>shareLinks[]</code> — every link ever created (token, URL, label, password, active, expiry); a legacy single <code>shareToken</code>
Closure and AAR	<code>archivedAt</code> , <code>closedBy</code> , <code>standDownReason</code> ; <code>aar</code> — four review answers and corrective actions

Taxonomy

The taxonomy lives in `client/src/crisis/taxonomy.ts`, mirrored on the server for write validation. There are 26 canonical types in 7 categories; two retired ids (`chemical` → HazMat, `security` → Violence / Threat) are accepted on write and aliased forward on read, and unknown ids fall back to Other. Default severity is advisory only; the record has no severity field.

Category	Types (default severity)
Natural Hazards	Wildfire (high), Hurricane / Tropical (high), Severe Weather (moderate), Winter Storm (moderate), Flood (high), Earthquake (high), Wildlife (low)
Fire & HazMat	Fire (Structure) (high), HazMat (high)
Infrastructure & Utilities	Utility / Power Outage (moderate), Water System (moderate), IT / Comms Outage (moderate), Cyber Incident (high)
Security	Violence / Threat (critical), Suspicious Activity (low), Theft / Loss Prevention (low), Civil Unrest (moderate)
Medical & Life Safety	Medical (moderate), Mass Casualty (critical), Fatality (critical), Search & Rescue (high), Public Health (moderate)
Access & Operations	Road Closure / Access (low), Maritime (moderate), Aviation (moderate)
Other	Other (low)

The crisis workspace tabs

An open incident shows a header, a tab strip, and on desktop a resizable dock (360 px to min(720 px, 50 vw)) whose upper 46% is a transparent window onto the live Cesium globe, with the Map Layers toolkit beneath.

Situation Report

The tab holds the Executive Summary, an Incident Information card (name, start and end, location, Property — including the fleet — type, and the status chips), the ICS org chart, the Actions & Events log and **Live Data Layers**, where the operator prescribes which live feeds appear on the share map, adds Property Pin groups, and attaches fleet vessels. The **Windstar Vessels** picker lists each hull with live AIS status polled every 2 minutes; changes are logged as `vessels-change` entries.

Action log

+ Action , + Event and + Info prepend a row stamped with the current time and the signed-in user. Images attach by picker or paste, are resized in the browser to at most 1200 px (JPEG 0.82) and uploaded unsigned to Cloudinary. Controls hide or show Info and System entries and switch between a paginated table (10 newest by default) and a **timeline** whose horizontal gaps encode elapsed time on a log scale saturating at one day.

The store also writes an audit entry for every state change, with `entryType: event` , the signed-in user as actor, a `system` kind and a `meta` payload. They are immutable on the server (`PATCH` returns 403). Twelve kinds exist:

Kind	Written when	Meta
created	Incident created	—
status-change	Status changed	from, to
complexity-change	ICS complexity changed (selector hidden)	from, to
assignment	Person assigned, including command transfer	roleId, roleTitle, name, replaced
assignment-ended	Assignment ended	roleId, roleTitle, name
role-added	ICS role added	roleTitle
role-removed	Role removed (cascades to sub-roles)	roleTitle, subRoles
share-created	Share link published	token, label
share-revoked	Share link revoked	token
stood-down	Stand-down completed	reason, releasedAssignments
reopened	Incident reopened	—
vessels-change	Vessels attached or removed	added, removed, count

System entries can still be deleted by any operator while the incident is live, so the trail is tamper-evident rather than tamper-proof.

ICS org chart

The chart seeds 21 built-in NIMS roles: Incident Commander at the root; Command Staff on dashed advisory connectors — Safety Officer, Public Information Officer (with a multi-person "GSOC Support" sub-role) and Liaison Officer; and four General Staff sections — Operations (Branch Director, Division/Group Supervisor), Planning (Resources, Situation, Documentation, Demobilization Unit Leaders), Logistics (Support and Service Branch Directors) and Finance/Admin (Time, Procurement, Compensation/Claims, Cost Unit Leaders). Built-in titles are locked; custom roles can be added, removed when unstaffed (cascading to sub-roles), restored or reset to defaults.

A **personnel pool** under the chart captures name, title, phone and email; drag-and-drop assigns a person to a role. One person cannot hold two roles at once; assigning to a non-support role ends the previous holder — a command transfer logged as " assigned as (replacing)"; support roles accumulate holders. The edit panel lists current holders and a scrollable **assignment history** with start → end stamps, the staffing record the after-action report reads.

Checklists

A role picker (IC, SO, PIO, LNO, OSC, PSC, LSC, FSC) selects a role card with a progress bar and three phases: **Immediate** (first minutes, highlighted red), **Ongoing** (operational period) and **Stabilization / Demobilization**. Every touched item shows "✓ Checked · " or "✗ Unchecked · " from server-stamped state.

One template exists in code: `vessel-grounding-v1` , with 90 items across 8 roles — Incident Commander 15, Safety 11, PIO 11, Liaison 10, Operations 13, Planning 10, Logistics 10, Finance 10. The per-type template hook is empty, so every type — a wildfire or a cyber incident included — receives the vessel-grounding checklist `partial` .

Intake

A first-notification questionnaire of 30 questions in six groups, also from the vessel-grounding template: A. Vessel Identification & Position (6), B. Life Safety & Casualties (5), C. Vessel Status & Stability (6), D. Environmental (4), E. Passengers & Guests (4), F. Command, Communications & Resources (5). The header counts "N/30 answered"; answers ride the incident record and build an intake table on the share link. The per-type hook is likewise empty `partial` .

Map layers and drawing tools

+ `New Layer` takes a name, a category type — Fire Perimeter, Burned Area, Flood Zone, Staging Area, Exclusion Zone, Search Grid, Other, each with a default colour — and a shape: Area (polygon, 3+ vertices), Line (2+), Arrow (a directional line toward the last vertex) or Point. Drawing happens on the full-screen globe with Undo / Finish / Cancel; on finish a thumbnail of up to 640×360 px is captured from the Cesium canvas and uploaded to Cloudinary.

Each layer row shows a measurement computed from its vertices (area and perimeter for polygons; length in km, miles and nautical miles, with bearing for a two-point run, for lines; lat/lon for points), Flip for arrows, visibility, Coords — which imports pasted decimal degrees, degrees with cardinal letters, DMS or WKT — and delete.

Vessels, share links and placeholders

Vessels are attached from the Situation Report and the header's **Share Links** panel is covered below. Two further tabs, **Resource Tracker** and **Comms Log**, render disabled with a "Coming soon" tooltip and have no store fields, components or endpoints behind them.

Multi-operator collaboration

Incidents are a shared workspace: every operator sees every peer's change within about a second. Three channels carry writes.

Channel	Data	Mechanism	Conflict behaviour
Incident blob	Summary, incident info, intake, org chart, personnel, draw layers, live-layer prescriptions, AAR	New ids <code>POST /api/incidents</code> immediately (server upserts); known ids <code>PUT /api/incidents/:id</code> debounced 1.5 s per incident. Log and checklists are omitted; the server preserves its own copies.	Last-write-wins; concurrent edits inside the window can overwrite each other (SSE narrows it to roughly 1 s).
Action log	Each entry	<code>POST /api/incidents/:id/log</code> (idempotent by id); <code>PATCH .../log/:entryId</code> after an 800 ms debounce not re-armed by further typing; <code>DELETE .../log/:entryId</code> . All under <code>SELECT ... FOR UPDATE</code> .	Conflict-safe per entry. A PATCH that 404s means a peer deleted the entry — delete wins.
Checklist toggles	Each item	Optimistic flip, then <code>POST /api/incidents/:id/checklist/:itemId</code> ; the server applies it under a row lock, stamps its own time and actor, and returns the authoritative map.	Conflict-safe per item; 500-item cap; 409 on archived incidents.

Every editor holds an EventSource on `/api/incidents/events` (ping every 25 s). The server broadcasts `upsert` and `delete` on every mutation from any route, including share-link viewers' checklist toggles. On receipt: if a local blob write is pending for that incident the event is dropped (the local write wins and its echo re-converges the client); otherwise the server's log and checklist map win except for entries and items still in flight locally; and if nothing differs the event is treated as the client's own echo. The stream has no replay, and fan-out is in-process memory.

The save indicator reads "Saving...", "All changes saved" (all three engines idle) or "Unsaved — will retry". When the tab is hidden or closing, pending blob writes are flushed with `navigator.sendBeacon` to `POST /api/incidents` (the `upsert` path stands in for the PUT and carries the auth cookie) and pending log edits with keepalive fetches.

Stakeholder sharing

A share link publishes a password-protected, read-mostly projection of one incident to people with no account. Any member can create one from the Share Links panel, with an optional audience label.

Generation. `POST /api/crisis/publish` mints a UUID token and a 10-character password from a 32-symbol alphabet with no `O/0/I/1`, stores `sha256(sha256(password))`, sets expiry to now + 72 hours, and returns the token, URL (`/?share=<token>`), password — once — and expiry. The panel lists links newest-first with label, status, expiry countdown (amber under 12 h), Renew (a fresh 72 h, not cumulative), Copy, the password chip, Revoke, and "Opened $N \times$ by $\sim M$ viewers · last " from an access log of token, IP and user agent, viewers being distinct IPs.

What is published. The projection strips personnel contact details (assignments reduce to id, role, name, start, end), sends vessel MMSIs only, excludes the personnel pool, share links and AAR. The open incident is re-published to every active link 1.5 s after any change, even with the overlay closed; stand-down and reopen publish explicitly.

The viewer's phone. A 401 shows a "Protected Situation Report" gate. The typed password is upper-cased, hashed in the browser with SHA-256 (HTTPS only) and sent as `?k=`, so the plaintext never travels in a URL. The page has three tabs:

- **Situation Report** — sticky header with a "last updated" readout that turns amber after 45 min without updates; executive summary; the intake table; a **Property Watch** card running the operator's FIRMS + NWS + USGS proximity scan (25-mile radius, polled every 5 min) filtered to the prescribed properties; **vessel cards** with live AIS status, position, destination and ETA polled every 2 min ("No position reported — not shown on the map" rather than an implied all-clear); Incident Details; the action log; the map; a Map Layers list; and a read-only ICS org chart.
- **IAP** — an in-page reader of the Incident Action Plan PDF for the incident type.
- **Checklists** — the editor's board, writable: viewers enter a name (default "Share viewer") and toggle items through `POST /api/crisis/share/:token/checklist/:itemId`, the only write a viewer can perform; the server applies it under a row lock and fans it to operators and every live snapshot. Archived incidents return 409.

Globe versus flat map. When the incident prescribes live layers, property groups or vessels, the page mounts a Cesium globe (72 vh) reusing the operator's layer components — drawn layers, property pins, vessel markers, tap-to-inspect shape cards, a measure tool, a Map Controls card (four basemaps, Zoom to Incident, Reset), one toggle chip per prescribed layer and legends where needed. With only drawn layers it renders a 420 px Leaflet map on Esri dark-grey raster tiles instead; pins and vessels are not drawn there. A Map Layers list lets viewers hide drawn layers but never surface one the team hid.

Live updates and closure. After unlocking, the page opens an EventSource on `/api/crisis/share/:token/events` (25 s keepalive; `update` on every publish or checklist change). A `revoked` event closes the stream and refetches, landing on the **closure page**: "Situation Report — Concluded", the incident name and final status, "Stood down" or "Report closed" with the timestamp, the final executive summary, and guidance to contact the GSOC; expired links show the same page.

The 18 shareable live layers. Only feeds served by public, keyless read paths are whitelisted; company asset, metered and cosmetic layers are excluded.

Group	Layers (provider)
Weather	Precipitation Radar (RainViewer), Precip Forecast (NOAA WPC QPF), Hurricanes (NHC + JTWC), Lightning (Blitzortung), Wind Particles (GFS), Wind Streamlines (GFS)
Fire & Smoke	Hotspots (NASA FIRMS VIIRS 24 h), Named Fires (NIFC), Smoke (NOAA HMS), Air Quality (AirNow + PurpleAir), 7-Day Fire Potential (NWCG), Fuel Models (LANDFIRE FBFM40)
Hazards & Alerts	NWS Alerts, Earthquakes (USGS M2.5+), Rivers & Floods (NOAA gauges), Power Outages (multi-state utility feeds)
Open-Source Intel	News (GDELT), Intel Feed (scanner / crime / social)

Incident Action Plan library

Each incident type can carry one Incident Action Plan PDF, plus one general default used when a type has none. Documents are stored as BYTEA in Postgres, one per type enforced by a unique index. On first boot the migration seeds the default, "Incident Action Plan — Generic Template", a 14-page ICS/NIMS fill-in-forms PDF whose metadata title reads "(Demonstration)".

Administrators manage the library from the Admin panel's IAP Library; upload and delete require the admin role. Uploads arrive as base64 JSON (25 MB body limit), are capped at 15 MB, must begin with the `%PDF-` magic bytes, and replace the existing document for that type. There is no version history and no per-incident IAP.

Share viewers read the document through `GET /api/crisis/share/:token/iap`, which resolves the snapshot's normalised type to its document or the default (404 `{noIap: true}` otherwise) and streams it inline with a 5-minute private cache. The IAP tab renders every page to canvas with pdf.js.

After-action reporting

The after-action report opens from the **PDF Report** button on an archived incident — only after stand-down; open incidents have no export path. It is a full-screen modal reading the live record: header, Executive Summary and Incident Details, Response Metrics, the ICS org chart fully expanded with the last holder of each seat, the ICS Progression swimlane, the Assignment Roster, the After-Action Review, Corrective Actions, the complete log with actors and inline images, an Incident Map and Map Layers list, and a timestamp footer.

Response Metrics — five cards computed purely from the captured record, covered by 17 unit tests:

Metric	Computation
Duration	(archivedAt, or now) – (incident start, or creation); "start → stand-down" or "start → now (still open)"
Time to Active	Creation → first status-change with to: active in the first activation run; "At creation" when created Active; "Never active" otherwise
Personnel	Unique people by pool id (fallback: normalised name), with " assignments · command transfers", a transfer being a consecutive different holder of any root role
Log entries	Total, with " operator · system"
Stakeholder reach	Access-log opens summed across every link the incident ever had, as "x" with "~ viewers · links"; "—" with no links

A footnote reports "ICS complexity at close: Type n" when set; spans are humanised as "18m", "5h 12m", "3d 4h".

ICS staffing swimlane — one row per role ever staffed, in chart order, with a bar per assignment clamped to the window from the earliest start or creation to stand-down (or now); open assignments are dashed. An SVG chart carries the legend "adjacent bars in one row = seat transfer". The **Assignment Roster** lists every assignment chronologically: name, role, assigned, released (or "active"), held for.

Four-question review — autosaving fields for "What was expected to happen?", "What actually happened?", "What went well, and why?" and "What can be improved, and how?", editable after archive and never shared. The **corrective-action tracker** adds rows with a done checkbox, text, owner and due date, with an "(n open)" counter.

Output path. "Print / Save as PDF" calls the browser's print dialog with an injected stylesheet: letter portrait, 0.6 in margins, dark theme flipped to paper-white, page-break protection on cards and headings, and a repeating page header. There is no server-side render, and no structured (CSV / JSON) export of AAR or checklist data exists **partial**.

Maturity at a glance

Capability	Status	Note
Crisis overlay, CRISIS button, tab-title badge	production	Badge counts Active, non-archived incidents
Incident dashboard and archive	production	Any member deletes live incidents; admins only for archived
Lifecycle statuses and audit entries	production	Transitions unenforced
Stand Down / Reopen sequences	production	Revocation failures non-blocking, reported inline
Incident record, Situation Report and taxonomy (26 types)	production	Severity advisory only; start/end stored without timezone
ICS complexity type	partial	Stored and audited; selector hidden
Action log, attachments, timeline	production	Unsigned upload to a hardcoded Cloudinary account; system entries deletable while live
ICS org chart, personnel pool, assignment history	production	Pool per-incident; role removal drops sub-role assignments silently
Checklist board and per-item sync	production	Server-stamped; 500-item cap
Checklist and intake templates	partial	One vessel-grounding template each (90 items, 30 questions) for all types
Map layers, drawing, coordinate import, measurements	production	MULTIPOLYGON imports collapse to one layer
Fleet vessels and Live Data Layers prescription	production	18 keyless layers; positions depend on server AIS configuration
Multi-operator sync	production	Blob last-write-wins (~1 s window); log and checklists row-locked and conflict-safe
SSE fan-out (editors and viewers)	production	In-process, single-server; no replay; no route tests
Share links (password, 72 h TTL, renew, revoke, access log)	production	No link ownership; key in query string; no rate limiting on public routes
Share page (three tabs, live map, viewer checklists, closure page)	production	Flat map omits pins and vessels; snapshot updates best-effort
IAP library	production	One PDF per type plus default; 15 MB cap; no versions
After-action metrics, swimlane, roster, review, corrective actions	production	Reach counts distinct IPs; exported only via the printed report

Capability	Status	Note
Print / Save as PDF	partial	Browser print only; no server-side render
Resource Tracker, Comms Log tabs	placeholder	Disabled "Coming soon" tabs; no data model
Operational periods / resource tracking model	placeholder	Not modelled; periods appear only as checklist phase labels

Data Sources

GSOC Monitor draws on roughly ninety distinct upstream feeds from about forty providers. Almost all of them are free public services; the exceptions are called out below. This chapter is the authoritative register of where every piece of data on the globe comes from, how it is reached, what it costs, and how fresh it is. "Path" says whether the operator's browser talks to the provider directly or the server proxies it; "Cadence" gives the effective refresh interval and any server cache.

Cost and key summary

Category	Sources	Notes
Free, no key	the large majority — NOAA/NWS, USGS, NASA, NIFC, NWCG, EPA AirNow data (key is free), CelesTrak, adsb.fi, Blitzortung, GDELT, Esri Living Atlas, Open-Meteo, MET Norway, OSM/Nominatim, RSS publishers, PulsePoint, CHP, Socrata, Bluesky, KUBRA/NISC utility maps	several require an identifying User-Agent by policy
Free, key required	AirNow, PurpleAir, aisstream.io, NPS Data API, Cesium ion	keys are free registrations; the AQI layer and ship AIS stream are empty without them
Freemium	Cloudinary image hosting	unsigned upload preset on the owner's free tier
Paid, optional	VesselFinder or MyShipTracking (vessel positions by IMO), Anthropic Claude (AI briefing), Google Map Tiles 3D	each is off unless its key is set
Paid, infrastructure	Heroku dyno (~\$5–7/month basic), Heroku Postgres add-on	the only unavoidable spend

Severe weather and environment

Source	Provider	Path	Auth	Cadence	Used by
Active alerts (<code>api.weather.gov/alerts/active</code>)	NOAA National Weather Service	browser-direct, server fallback	User-Agent policy	60 s poll; server cache 60 s stale-on-error	Alerts layer, Property Watch, dashboard, risk report, screensaver
County polygons by FIPS	US Census via plotly/datasets on GitHub	browser-direct	none	loaded once per session	resolves county-coded alerts to polygons
Radar and IR satellite tiles	RainViewer	manifest via server, tiles browser-direct	none	manifest 2 min; tiles browser-cached	Precipitation Radar layer
QPF precipitation forecast (24/48/72 h)	NOAA Weather Prediction Center MapServer	browser-direct	none	per risk report; layer rasters on toggle	Precip Forecast layer, risk report
Global wind grid (5°, 2,376 points)	Open-Meteo (NOAA GFS)	server	none	30 min background refresh → Postgres snapshot → baked fallback	Wind particles, streamlines, arrows

Source	Provider	Path	Auth	Cadence	Used by
Point wind forecast (7-day hourly)	Open-Meteo, MET Norway fallback	server	none / User-Agent	30 min cache; 10-min circuit breaker to met.no	Wind probe panel, risk report
10-day daily forecast	Open-Meteo	server	none	30 min cache	Risk report forecast strip
Dashboard weather (current temp/wind, 7-day precip)	Open-Meteo	browser-direct	none	one batched call per 5-min dashboard scan	Property Status Dashboard
Daily MODIS true-colour imagery	NASA GIBS (Terra and Aqua)	browser-direct	none	immutable per date	Earth basemap, risk report smoke map

Wildfire, smoke and air quality

Source	Provider	Path	Auth	Cadence	Used by
VIIRS thermal hotspots (24 h, FRP)	NASA FIRMS via Esri Living Atlas	browser-direct	none	5 min plus camera moves; 2,500 cap	Fires layer, Property Watch, dashboard, risk report
Current wildfire incidents (≥5 acres, <100 % contained)	NIFC WFIGS / IRWIN	browser-direct	none	5 min; 800 cap	Named Fires layer, risk report
Current fire perimeters (≥100 acres)	NIFC WFIGS	browser-direct	none	5 min; 500 cap	Named Fires layer
7-day significant fire potential (PSA polygons)	NWCG Predictive Services	server	none	2 h cache; client 30 min	Fire Outlook layer, risk report
Smoke plumes (daily KML)	NOAA NESDIS Hazard Mapping System	server	none	2 h cache; tries today then two days back	Smoke layer, risk report
Reference monitors (PM2.5, O3, PM10, CO, NO2, SO2)	US EPA AirNow	server	<code>AIRNOW_API_KEY</code> (free)	15 min shared cache	AQI layer, dashboard
Community PM2.5 sensors	PurpleAir	server	<code>PURPLEAIR_API_KEY</code> (free read key)	15 min shared cache; EPA correction applied; 6,000-sensor cap	AQI layer
Fuel model raster (LF2024 FBFM40, 30 m)	USGS/USFS LANDFIRE ImageServer	browser-direct	none	per zone / per report; CONUS only	Fuel layer, Fuel Analyzer, risk report

Seismic, tropical, lightning and flood

Source	Provider	Path	Auth	Cadence	Used by
Earthquake summary feeds (hour/day/week × magnitude)	USGS Earthquake Hazards Program	server	none	60 s cache; client 1–15 min by period	Earthquakes layer, Property Watch, dashboard

Source	Provider	Path	Auth	Cadence	Used by
Active hurricanes (positions, tracks, cones)	NOAA NHC via Esri Living Atlas	browser-direct	none	5 min, five sublayer queries	Hurricanes layer, dashboard, share globe
Tropical weather outlook (2/7-day development regions)	NOAA NWS tropical MapServer	browser-direct	none	5 min	Hurricanes layer
Invest advisories (West Pacific, Indian Ocean, South Pacific)	US Navy Joint Typhoon Warning Center	server	none	1 h cache, text bulletins parsed	Hurricanes layer, dashboard
Live lightning strikes	Blitzortung.org community relays (unofficial WebSocket)	browser-direct and server	none	streaming; 10-min / 2,500-strike window per viewer	Lightning layer, ticker
Lightning history (1–24 h)	Blitzortung via server collector	server	none	1-in-6 decimated ring buffer, Postgres chunks every 5 min; client 30 s	Lightning History layer, risk report
River gauges (~12.7k) and gauge detail	NOAA National Water Prediction Service	server	User-Agent policy	15 min background refresh; detail 10 min cache	Rivers & Floods layer

Infrastructure, aviation, maritime and space

Source	Provider	Path	Auth	Cadence	Used by
Statewide and utility outage feeds (5 ArcGIS services)	Cal OES, Salt River Project, SSVEC, Minnesota Power, Riverside PU	server	none	4-min aggregate rebuild	Outages layer
Storm Center outage maps (9 utilities: Oncor, Georgia Power, JEA, Colorado Springs, Cobb EMC, LG&E/KU, Evergy, Versant, Appalachian Power)	KUBRA	server	none	4 min; quadkey tile walk capped at 150 tiles / 25 s	Outages layer
Co-op outage maps (Sawnee, SLEMCO, Price Electric, Cloverland)	NISC	server	none	4 min	Outages layer
ADS-B positions for ten tracked registrations	adsb.fi open data	server	User-Agent; ~1 req/s guidance honoured	10 s active / 60 s idle; Postgres snapshot	Flights layer
Airframe registry (make, model, operator)	adsbdb.com	server	none	7-day refresh	Flight detail panel
Aircraft photos	Planespotters.net	server (thumbnails hotlinked)	none, attribution shown	7-day refresh	Flight detail panel
Live AIS (fleet MMSIs)	aisstream.io	server WebSocket	AISSTREAM_API_KEY (free)	streaming; community receivers only	Ships layer

Source	Provider	Path	Auth	Cadence	Used by
Fleet last-known positions	CruiseMapper (HTML scrape)	server	none; disable with env	every 120 min	Ships layer (default position source)
Fleet positions by IMO	VesselFinder or MyShipTracking	server	paid key	every 120 min	Ships layer (when configured)
Reverse geocoding for snapshots and cards	BigDataCloud	browser-direct	none	per click	Fleet snapshot, screensaver cards
Coastlines and major cities	Natural Earth, GeoNames (bundled)	bundled	none	static	Fleet snapshot map
Two-line element sets (stations, brightest, GPS, weather, Starlink)	CelesTrak	server	none	2 h cache per group; propagation client-side	Satellites layer, ISS tour

News and open-source intelligence

Source	Provider	Path	Auth	Cadence	Used by
World news RSS (BBC, Guardian, Sky, NPR, Al Jazeera)	publishers	server	none	5 min cache, 120-story cap	Breaking News widget, ticker
User-added RSS feeds	any publisher	server	none	5 min per feed	Breaking News (per-browser)
Park alerts and news releases (six parks)	US National Park Service Data API	server	NPS_API_KEY (free; DEMO_KEY default)	10 min	News ticker Park mode
Geolocated security news events	GDELT GKG GeoJSON v1	server	none	10 min cache per query	News map layer (share pages), dashboard
Google News RSS search	Google News	server ingest	none	90 s	Intel Feed
Bluesky author feeds and keyword search	Bluesky (AT Protocol)	server ingest	none	90 s	Intel Feed
Fire/EMS dispatch incidents	PulsePoint (undocumented web-app endpoint)	server ingest	none	90 s	Intel Feed
Statewide highway dispatch log	California Highway Patrol	server ingest	none	90 s	Intel Feed
Open-data crime datasets (built-in: Chicago)	Socrata portals	server ingest	none	90 s	Intel Feed
Generic RSS/Atom watchlist sources	any public publisher	server ingest	none; SSRF-validated	90 s	Intel Feed

Source	Provider	Path	Auth	Cadence	Used by
Place geocoding for intel items	OpenStreetMap Nominatim	server	User-Agent policy	30-day cache; 16 lookups per cycle	Intel Feed, watchlist pins
Place search	OpenStreetMap Nominatim	server	User-Agent policy	24 h cache per query	Search bar
County and park boundaries	Esri Living Atlas, NPS Land Resources Division	server	none	24 h cache	National Parks screensaver
Nearby hospitals, hotels, police, fire stations; driving routes	Overpass API, OSRM demo router	browser-direct	none	on demand	Location detail panel

Basemaps, platform and AI

Source	Provider	Path	Auth	Cadence	Used by
Gray canvas, imagery and label tiles	Esri ArcGIS Online	browser-direct	none	browser cache	Dark, Light, Dark Sat and Earth basemaps
Topographic tiles	OpenTopoMap	browser-direct	none	browser cache	Topographic basemap
World Terrain and OSM Buildings	Cesium ion	browser-direct	VITE_CESIUM_ION_TOKEN (free tier; demo token fallback)	tileset cache 192 MB	3D Buildings & Terrain
Photorealistic 3D Tiles	Google Map Tiles API	browser-direct	VITE_GOOGLE_MAPS_KEY (paid, metered)	—	hidden layer
OpenStreetMap raster tiles	OSM Foundation	browser-direct	none	browser cache	screensaver minimaps
Image hosting for incident photos and map thumbnails	Cloudinary (unsigned preset)	browser-direct	none	—	Crisis action log, drawn layers
AI duty-officer briefing	Anthropic Messages API (Claude)	server	ANTHROPIC_API_KEY (paid)	10-min cache by payload hash; rules digest fallback	Property Status Dashboard
Web fonts (Inter, JetBrains Mono)	Google Fonts	browser-direct	none	browser cache	UI typography
Postgres	Heroku Postgres	server	DATABASE_URL (paid add-on)	—	users, incidents, share links, watchlist, lightning history, snapshots, IAP PDFs

Provider policies worth knowing

- **Identifying User-Agent.** NWS, Nominatim, CelesTrak and adsb.fi ask callers to identify themselves. The server sends `NWS_USER_AGENT` to all of them; when the variable is unset the default string carries no contact address, which does not satisfy the NWS or Nominatim policies.
- **Polling limits.** adsb.fi asks for roughly one request per second (honoured with sequential 150 ms spacing), CelesTrak asks for TLEs to be cached for hours (2-hour cache, honoured), Nominatim allows about one request per

second (24-hour search cache; intel geocoding budgeted at 16 lookups per 90-second cycle with 1.1-second spacing), and GDELT rate-limits shared Heroku IP addresses (10-minute cache and in-flight coalescing).

- **Unofficial sources.** Three feeds have no published API contract: the Blitzortung relays (used the way the public lightningmaps.org client uses them), the PulsePoint web-app endpoint (decrypted with a static passphrase recovered from its public client), and the CruiseMapper ship pages (HTML scraping behind Cloudflare). Each can stop working without notice, and each is explicitly flagged as fragile in the code.
- **Geographic coverage.** LANDFIRE fuels and AirNow/PurpleAir AQI are CONUS-only; HMS smoke covers North America; NWS alerts, NWPS gauges, NIFC incidents and the outage feeds are US-only; FIRMS, NHC/JTWC, earthquakes, lightning, wind, flights, ships, satellites and GDELT are global.

Capability Maturity and the MVP Feature Set

The code review catalogued 303 distinct capabilities across twelve subsystems and rated each one by the state of its implementation. The result is a platform that is far more complete than a typical minimum viable product: 253 capabilities (84%) are production-ready, 34 are working but rough, 13 are partial, and 3 are scaffolds with no function behind them.

303

CAPABILITIES

84%

PRODUCTION-READY

~90

DATA FEEDS

1

DYNO TO RUN IT

Maturity by subsystem

Subsystem	Production	Rough	Partial	Scaffold	What is not production-grade
Globe shell and chrome	29	5	2	—	search has no typeahead; render-quality controls are dead code; hidden Google 3D layer; login-screen telemetry is decorative
Hazard layers (alerts, quakes, fire, smoke, AQI, outlook)	21	—	2	—	earthquake period filter has no UI; six of seven layers lack an on-map legend
Movement layers (flights, satellites, ships, hurricanes, lightning, rivers, outages)	29	3	—	—	ship positions depend on a fragile scrape; hurricane feed has no server cache; satellites never re-fetch TLEs while on
Environmental layers (radar, precip, wind, fuel, 3D, time zones, pins)	17	4	4	—	radar recolour pipeline is a workaround; IR "Combined" mode uncalibrated; directions panel uses public demo servers; news/intel pins hidden from the operator app
Analytics (dashboard, threat score, Property Watch, risk report, fuel analyzer)	14	6	—	—	risk report is wildfire-only and print-only; property list is hardcoded; no background monitoring or alerting
Crisis lifecycle (incident record, tabs, sync, stand-down)	36	1	2	1	one checklist and intake template for all 26 incident types; Resource Tracker and Comms Log are placeholders; complexity selector hidden
Crisis sharing and reporting (share links, viewer, AAR)	34	2	—	—	AAR and reports are browser-print only; AAR only after archive
Server core (auth, admin, cache, boot, deploy)	14	5	1	—	no token revocation or role management; proxy-unaware rate limit; required env vars undocumented; unit tests only
Server crisis API	22	1	—	—	no rate limiting on public share or briefing routes
Server hazard routes	12	1	—	—	temporary radar diagnostic endpoint still shipped
Server movement and geography routes	17	2	—	2	two dead routes (directions, drive); ship sources fragile
Server intel and news	8	4	2	—	news geolocation is a 95-entry gazetteer; intel buffer is ephemeral; no alerting

The recommended MVP

The MVP question for this product is not "what must be built" but "what should be switched on, hardened and supported for a first customer". The recommendation below keeps everything that is production-ready and load-bearing, flags what ships with a caveat, and defers what is fragile or half-wired.

Core — ship and support

- **Globe and shell:** five basemaps including daily MODIS imagery, dockable panels with the mobile deck, sidebar, legends, provenance popover, search, measure tool, screenshot and fullscreen.
- **Hazard monitoring:** NWS alerts, USGS earthquakes, FIRMS hotspots, NIFC named fires and perimeters, HMS smoke, NWCG 7-day fire potential, RainViewer radar with timeline, WPC precipitation forecast, GFS wind (particles, streamlines, probe), NHC/JTWC tropical, Blitzortung lightning (live and 24-hour history), NWPS river gauges, multi-state power outages.
- **Air quality** with AirNow and PurpleAir — provided the two free keys are provisioned, since the layer is empty without them.
- **Asset awareness:** property pins, the ADS-B roster (company aircraft, hurricane hunters, fire tankers) with trails and takeoff/landing events, the satellite tracker and ISS mode.
- **Decision support:** Property Watch proximity scan, the Property Status Dashboard with its explainable threat score and live feed, the duty-officer briefing (rules digest by default; Claude when a key is set), the Property Wildfire Risk Report, and the LANDFIRE fuel-zone analyzer.
- **Crisis response:** the full incident workspace — situation report, action log with system events, ICS org chart, checklists, intake, map drawing, vessel attachment — with multi-operator live sync, share links with password, expiry, revocation and access log, the IAP library, stand-down and reopen, and the after-action report.
- **Platform:** invite-code authentication, admin panel, Heroku deployment, CI typecheck/test/build.

Included with a stated caveat

- **Fleet vessel tracking.** The default free path (community AIS plus a CruiseMapper scrape) sees the fleet intermittently and can be blocked without notice; a paid VesselFinder or MyShipTracking key is the only dependable source and should be budgeted (the code estimates roughly €85/month at two-hour polling).
- **Breaking News and the Intel Feed.** Both work today, but severity and category are keyword heuristics, the intel buffer empties on every restart, and two of the built-in sources (PulsePoint, CHP) have no API contract. Position them as beta.
- **Screensaver tours.** Production-quality and valuable on wall displays, but non-essential; no idle auto-start exists.
- **Location details panel** (nearest hospital, hotel, police, fire; drive routes). Depends on volunteer Overpass and OSRM demo servers called from the browser; results are labelled as estimates.

Defer past the MVP

- The hidden Google Photorealistic 3D Tiles layer (metered API, key would be baked into the bundle).
- The GDELT news map and intel pins on the operator globe (currently reachable only on share pages).
- Resource Tracker and Comms Log tabs (placeholders), the ICS complexity selector (hidden), per-type checklist and intake templates (only the vessel-grounding template exists).
- Dead code that should be removed before release: the `/api/directions` and `/api/drive` routes, the viewport-flights client path, the render-quality store, and the temporary radar diagnostic endpoint.

MVP readiness gates

Feature scope is not the constraint on shipping; the security posture is. The companion bug report lists the findings that should be closed before the MVP is exposed to a customer. The most important are session revocation on user deletion, the share-link password gate accepting the stored hash as a key, plaintext share passwords returned to every signed-in user, the unauthenticated briefing route that can drive paid model calls, the news endpoint that fetches arbitrary URLs, the proxy-unaware login rate limiter, the context-loss rebuild that throws in every layer's cleanup, and documenting the two required environment variables so a fresh deployment does not fail silently. None of them is architecturally difficult; together they are days of work, not months.

Known Limitations

These are the boundaries of the platform as built. They are not defects — the bug report covers those — but they shape what the product can promise today and where the roadmap should go.

Single-tenant by construction

The monitored properties (31 locations in 13 groups), the seven-ship fleet, the ten tracked aircraft, the checklist and intake templates, the six national parks in the news ticker and the AI briefing's system prompt are all constants in source code. Adding a property or a vessel means a code change and a redeploy. There is no property register, no asset database and no tenant boundary; the same fleet roster is duplicated in three places that must be kept in sync by hand.

Single-process by design

The shared cache, the authentication failure counter, both Server-Sent Events fan-outs, the intel buffer and every background collector live in the memory of one Node process. The code says so explicitly and names Redis as the prerequisite for a second dyno. Until then, horizontal scaling is not available, and every restart empties the caches and the intel feed.

Monitoring is pull-only

Nothing in the platform pushes. The dashboard scans only while it is open, threat scores have no history, the intel engine has no keyword or geofence alerts, and there is no email, SMS, Slack or push notification path. An operator must be looking at the screen to learn that a property's status changed.

Coverage is US-centric

LANDFIRE fuels and both air-quality sources are CONUS-only; HMS smoke covers North America; NWS alerts, NWPS gauges, NIFC incidents and the outage feeds are US-only, and the outage aggregator covers about nineteen states from hand-picked utilities. The wildfire risk report therefore degrades to "unavailable" sections for international assets, and the fleet — which sails internationally — is covered only by global feeds (tropical cyclones, lightning, wind, earthquakes, GDELT news).

Analysis depth

The threat score is an additive, uncalibrated heuristic by design. The property risk report implements one hazard (wildfire) and explicitly omits relative humidity, fuel moisture and terrain slope. The fuel-zone score uses app-defined weights rather than published fire-behaviour outputs and does not adjust for live wind or moisture. The wind grid is five degrees, too coarse for isobars or further environmental fields. Several dashboard signals evaluate only each group's first location.

Reporting is browser-print

Every report — the after-action report, the property risk report, the incident archive — is produced through the browser's print dialog with a print stylesheet. There is no server-side PDF engine, no persisted report, and no structured export of AAR data, checklist history or the access log.

Crisis workflow gaps

One checklist template and one intake questionnaire serve all 26 incident types. There is no operational-period object, no resource tracking and no communications log. Lifecycle transitions are unenforced (an operator can jump from Monitoring to Closed), system audit entries can be deleted while an incident is live, the personnel pool is re-entered per incident, and blob fields (executive summary, org chart, drawings) remain last-write-wins between concurrent operators.

Identity and administration

Accounts are gated by one shared signup code. There is no password reset, email verification, multi-factor authentication, role promotion or account disablement; the only admin actions are listing users, deleting them and rotating the code. Sessions are 30-day tokens verified by signature alone.

Operational transparency

Two required environment variables and eight optional keys are not documented in the repository's example file or Heroku manifest; a deployment created from the manifest alone fails every authenticated call. README statements about server-side alert geometry and viewport-based flight refresh describe code that no longer exists.

Test coverage

Twenty-two unit-test files cover pure logic — sync canonicalisation, checklist merges, AAR metrics, measurement math, the flight state machine, the cache. No test exercises an HTTP route, authentication, the migration, a map layer component or the intel adapters, and CI has no database, so the DB-backed code paths run untested.